QA evidence — migration 004 (delta_artifacts)

Field	Value
Run ID	20260608-migration-004
Date (UTC)	2026-06-08T14:03Z
Engine	PostgreSQL 16.14 (`docker.io/library/postgres:16-alpine`, podman 5.8.2)
Container	`helix-mig004-test` on `127.0.0.1:55446` (removed after run)
Migrations under test	`004_delta_updates.{up,down}.sql`
Applied stack	`001_initial_schema` -> `002_staged_rollout` -> `004_delta_updates`
Evidence	`run.log` — full real psql transcript

What `004` adds

A new `helix_ota.delta_artifacts` table (purely additive — no `ALTER` of any 001 table; `artifacts.artifact_type` already permits `'delta'`). Per `delta_updates_design.md` §4.1(a): a separate table holding the base->target relationship, off the `artifacts` row.

- `base_artifact_id`, `target_artifact_id`, `delta_artifact_id` — all `UUID -> helix_ota.artifacts(id) ON DELETE CASCADE` (design §4.2). `delta_artifact_id` nullable (NULL until the job publishes the payload).
- `CHECK (base_artifact_id <> target_artifact_id)` — `delta_artifacts_base_ne_target_chk`.
- `UNIQUE (base_artifact_id, target_artifact_id)` — one relationship row per pair.
- Delta file integrity, mirroring 001's `artifacts` conventions: `file_path VARCHAR(500)`, `file_size BIGINT` (`> 0` when present), `checksum_sha256 CHAR(64)` (`~ '^[0-9a-f]{64}$'` when present).
- `status` lifecycle CHECK (`pending|generating|generated|failed|cancelled`, design §3.3), `savings_percent` (0..100, NULL until measured — design §7/§9 anti-bluff), `partition_deltas`/`generation_errors` JSONB.
- Indexes for "find a delta from base X to target Y": the `UNIQUE(base,target)` backs the exact-pair lookup; `idx_delta_artifacts_base` and `idx_delta_artifacts_target` back the single-leg fan-out queries the selection matrix runs at update-check time; plus

idx_delta_artifacts_status (servable filter) and
idx_delta_artifacts_delta (payload back-join).

Validation result (real captured evidence — see run.log)

Check	Outcome
001 up	OK
002 up	OK
004 up	OK (table + 4 indexes + 5 CHECKs + 3 FKs present, confirmed via \d)
Seed base + target full artifacts	OK (2 rows)
Insert valid delta 1.0.0 -> 1.0.1 (generated, 90.23%)	OK
"find delta from base X to target Y" indexed lookup	OK (1 row)
base == target insert	**REJECTED** — delta_artifacts_base_ne_target_chk
Duplicate (base,target) insert	REJECTED — delta_artifacts_base_target_uniq
Dangling base FK insert	REJECTED — delta_artifacts_base_fk
ON DELETE CASCADE (delete base artifact)	OK — delta row count 1 -> 0
004 down	OK — delta_artifacts gone; artifacts/rollouts intact
002 down	OK
001 down	OK — helix_ota schema fully removed

No bluff: every line above is a real psql -v ON_ERROR_STOP=1 result captured in run.log against a live Postgres 16 instance. The container was removed (podman rm -f helix-mig004-test) after the run.

Reproduce

```
podman run -d --name helix-mig004-test -e POSTGRES_USER=helix \
  -e POSTGRES_PASSWORD=helix -e POSTGRES_DB=helix_ota \
  -p 55446:5432 docker.io/library/postgres:16-alpine
# wait for ready, then with:
#   PGPASSWORD=helix psql -h 127.0.0.1 -p 55446 -U helix -d
        helix_ota -v ON_ERROR_STOP=1
# apply 001 up -> 002 up -> 004 up; exercise inserts; 004 down ->
        002 down -> 001 down.
podman rm -f helix-mig004-test
```